Country-specific SCCs + Exam 2 review

ECON-4210

Economics of Climate Change

Prof. Casey Wichman

Georgia Institute of Technology

Today's class

- Ghost emissions
- Country-level social cost of carbon
- Exam 2: format and coverage
- Problem set 2 questions?
- Practice problems

nature climate change

ARTICLES

https://doi.org/10.1038/s41558-018-0282-y

Corrected: Author Correction

Country-level social cost of carbon

Katharine Ricke[1,2*], Laurent Drouet[3], Ken Caldeira[4] and Massimo Tavoni[3,5]

The social cost of carbon (SCC) is a commonly employed metric of the expected economic damages from carbon dioxide (CO_2) emissions. Although useful in an optimal policy context, a world-level approach obscures the heterogeneous geography of climate damage and vast differences in country-level contributions to the global SCC, as well as climate and socio-economic uncertainties, which are larger at the regional level. Here we estimate country-level contributions to the SCC using recent climate model projections, empirical climate-driven economic damage estimations and socio-economic projections. Central specifications show high global SCC values (median, US$417 per tonne of CO_2 (tCO_2); 66% confidence intervals, US$177–805 per tCO_2) and a country-level SCC that is unequally distributed. However, the relative ranking of countries is robust to different specifications: countries that incur large fractions of the global cost consistently include India, China, Saudi Arabia and the United States.

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Estimating country-specific SCCs

- Why?
  - To highlight economic damages that occur to each country.

- The global SCC is the “correct” concept to use for a global externality, but disaggregating the SCC to individual countries allows for better understanding of regional impacts and better facilitate international cooperation.

- Ricke et al. (2018) develop a “modular” IAM focused on estimating the SCC for each country (CSCCs) using the Burke, Hsiang, Miguel (2015) damage function
  - Uses Ramsey discounting framework to adjust for country-specific growth rates
  - For each country, they simulate GDP through 2100 once with an extra ton of CO_2 and once without, using that country’s projected warming and the Burke et al. temperature–growth curve.

Median global SCC estimate: $417 ($\rho = 2\%; \mu = 1.5$)

Note: $r = 2\% + 1.5 \times 2\% = 5\%$

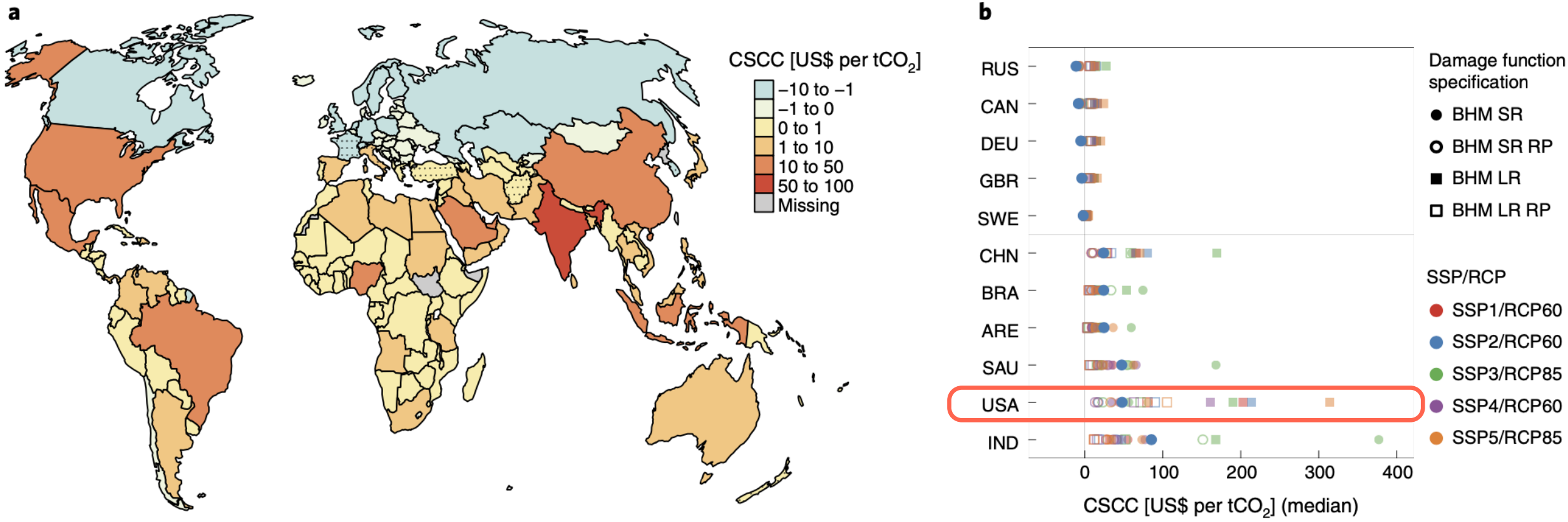
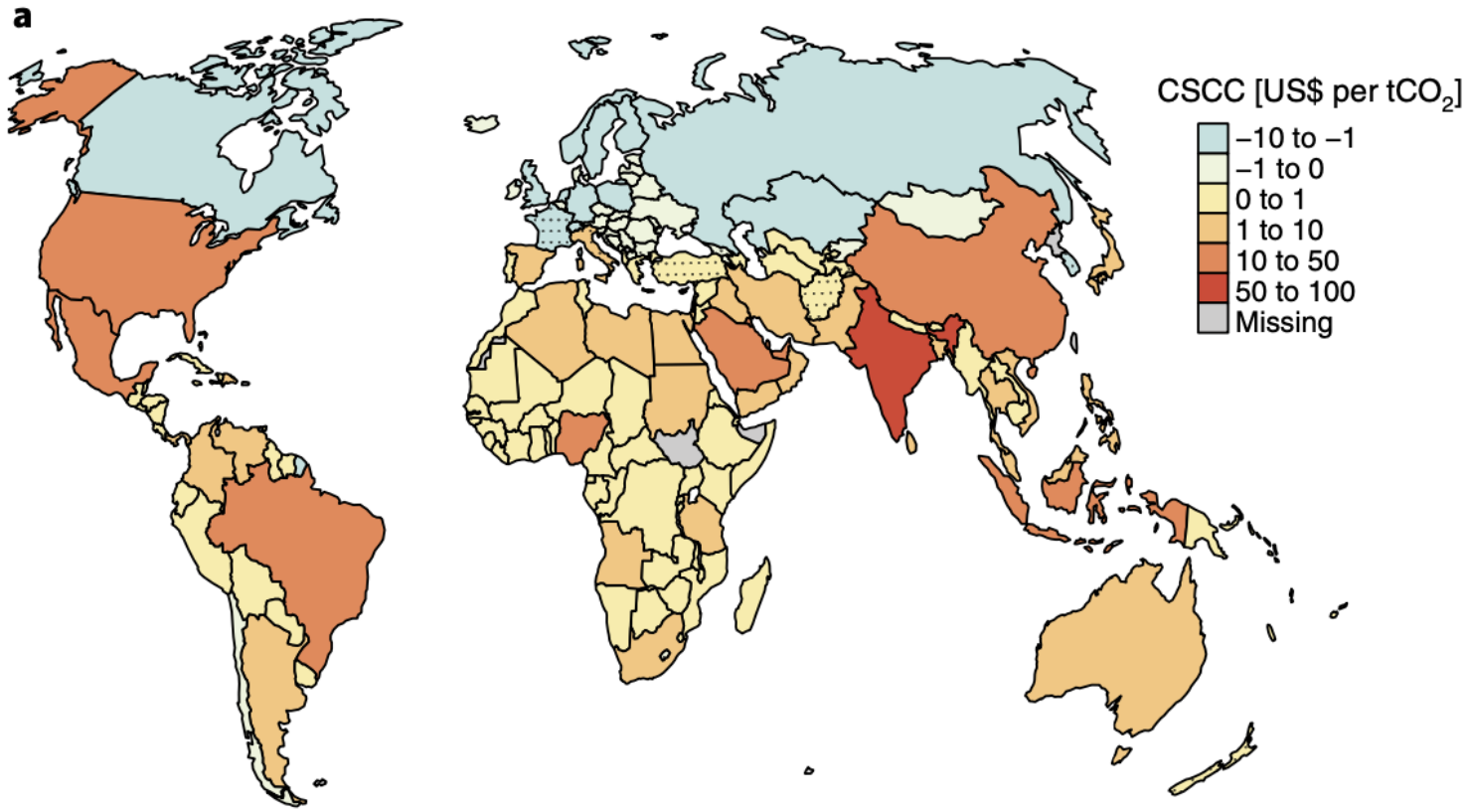


Fig. 2 | CSCCs. a, Spatial distribution of median estimates of the CSCCs computed for the reference case of scenario SSP2/RCP6.0, BHM-SR and a growth-adjusted discount rate ($\rho = 2\%$, $\mu = 1.5$) . Stippling indicates countries in which BHM damage function is not statistically robust[30]. **b**, CSCCs for alternative scenarios and damage function specification combinations for the five smallest and six largest CSCCs in the reference case (blue open circles). RUS, Russia; CAN, Canada; DEU, Germany, GBR, Great Britain; SWE, Sweden; CHN, China; BRA, Brazil; ARE, United Arab Emirates; SAU, Saudi Arabia; USA, United States; IND, India.

Ricke et al. (2018), https://www.nature.com/articles/s41558-018-0282-y

Median global SCC estimate: $417 ($\rho = 2\%; \mu = 1.5$)

Note: $r = 2\% + 1.5 \times 2\% = 5\%$



- India — $86/ton (21% of GSCC)
- US — $48/ton (11%)
- Saudi Arabia — $47/ton (11%)
- Brazil — $24/ton
- China — $24/ton
- UAE — $24/ton

- Several northern countries have *negative* CSCCs.

Fig. 2 | CSCCs. a, Spatial distribution of median estimates of the CSCCs computed for the reference case of scenario SSP2/RCP6.0, BHM-SR and a growth-adjusted discount rate ($\rho = 2\%$, $\mu = 1.5$) . Stippling indicates countries in which BHM damage function is not statistically robust[30]. **b**, CSCCs for alternative scenarios and damage function specification combinations for the five smallest and six largest CSCCs in the reference case (blue open circles). RUS, Russia; CAN, Canada; DEU, Germany, GBR, Great Britain; SWE, Sweden; CHN, China; BRA, Brazil; ARE, United Arab Emirates; SAU, Saudi Arabia; USA, United States; IND, India.

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Global inequality in climate damages

- **Lorenz curves:** measure the degree of income inequality (e.g., X% of the population have Y% of the wealth). A 45-degree line would be a perfectly equal income distribution.
  - Here, the Lorenz curve for CSCCs show that X% of the global population would experience what level of climate damages.
  - Countries ordered by CSCC per capita.

- **Gini coefficients:** summarize income inequality.
  - Ratio defined in graph by:
    - A/(A+B).
  - Numbers closer to 1 are bad.

Ricke et al. (2018), https://www.nature.com/articles/s41558-018-0282-y

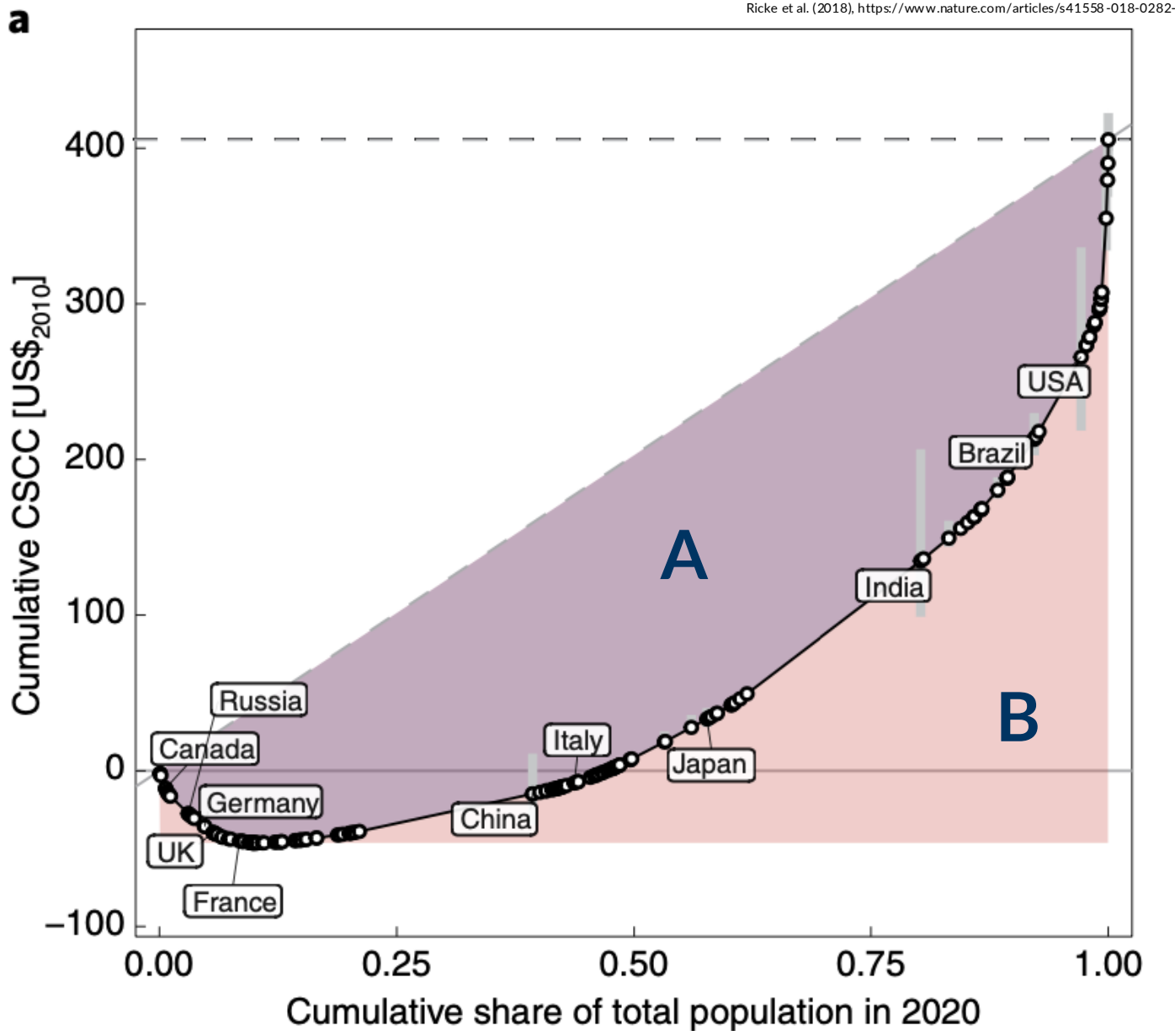


Fig. 3 | Lorenz curve and Gini coefficients for the country-level contributions to the GSCC in 2020. a, Cumulative global population plotted against cumulative SCC, with countries ranked by CSCC per capita, produces a Lorenz curve for the reference case of scenario SSP2/RCP60, BHM-SR and a growth adjusted discount rate with $\rho = 2\%$ and $\mu = 1.5$. The red and purple areas illustrate the quantities required to calculate the Gini coefficient, a synthetic metric of heterogeneity, or inequality, which is equal to the purple area divided by the sum of the purple and red areas. **b**, Gini coefficients for four damage model specifications. SSPs are distinguished by colour for both three fixed discounting rates; 2.5%, 3% and 5%), and four growth-adjuste discounting with ρ=(1%,2%) and μ=(0.7,1.5). The reference case illustrated in **a** (Gini coefficient, 0.62) is shown with a large solid blue circle.

“Winners” and “Losers”

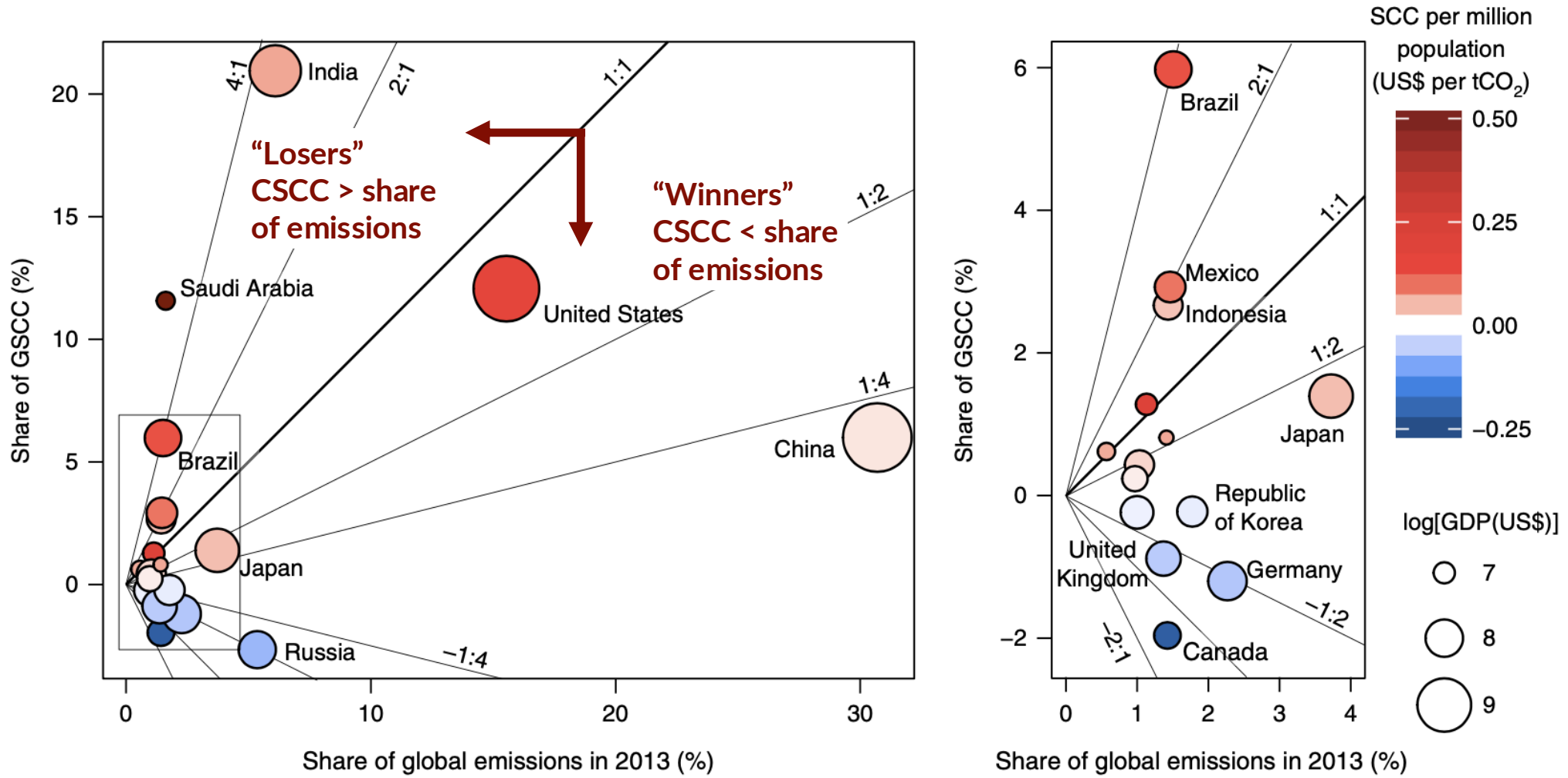
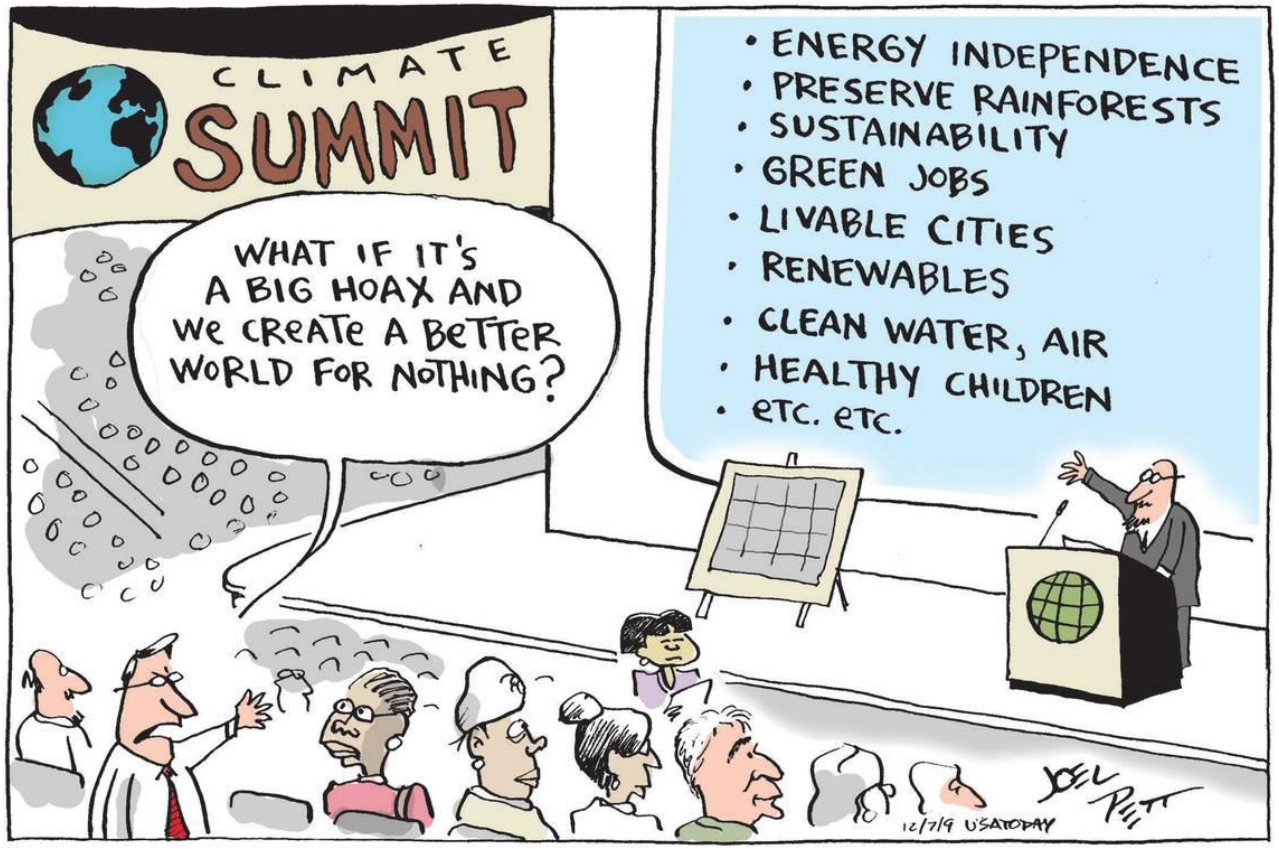


Fig. 4 | Winners and Losers of climate change among the G20 nations. Country-level shares of the GSCC (that is, CSCC/GSCC) versus shares of the 2013 CO2 emissions. The CSCC is the median estimate with growth adjusted discounting ($\rho = 2\%$, $\mu = 1.5$) for SSP2/RCP6.0 and BHM-SR. Points are shaded by CSCC per capita and sized according to country GDP in 2015. Diagonal lines show the ratio of the GSCC share to the emissions share. Ratios greater than 1:1 indicate that a country's share of the GSCC exceeds its share of global emission. The box in the left panel indicates the bounds of the detail shown in the right panel.

Ricke et al. (2018), https://www.nature.com/articles/s41558-018-0282-y

Key takeaways on country-specific SCCs

- Climate damages are heterogeneous
  - Even with CSCCs, the US CSCC is much greater than what the "domestic-only" SCC from the Trump admin because of the updated damage function.
- Some countries may be better off with climate change (represented by a negative CSCC), which may make international cooperation difficult.
- Some countries have high CSCCs, but small contributions to global GHG emissions.
- The CSCC highlights extreme global inequality in damages from climate change.



Exam 2

Monday, October 19, in class

- 20 multiple choice questions, 1 point each.
- 2 short answer questions, 10 points each.
  - Show your work. Partial credit for a correct method with an arithmetic error.
- One-page cheat sheet. No other references, books, or technology.
- You may use a calculator.

What the exam covers

Classes 9–13, today's material, and Problem Set 2

- 09 — (How much) should I care about my grandchildren's future?
- 10 — Measuring and valuing damages: damage functions
- 11 — Measuring and valuing damages: GDP and agriculture
- 12 — Heat-related mortality and valuing mortality risk
- 13 — Social cost of carbon
- Today — Country-level social cost of carbon
  - Readings we've covered in class are fair game.

Practice problem 1

Discounting and the Ramsey equation

- Sea-level rise and heavier storms will flood a coastal neighborhood more often. The state can buyout/relocate the homes for $4B today.
- The buyout avoids $15B of flood damages in year 40 and $40B in year 80.

  - (a) What is present value of the avoided damages at 7%
  - (b) What is present value of the avoided damages at 3%
  - (c) What is the NPV at each rate? Should the state buy out the homes?
  - (d) What if we used a Ramsey rate with $\delta = 1\%$, $\eta = 1$, $g = 2\%$? What if future generations' welfare gets equal weight ($\delta = 0\%$)?

Practice problem 1

Avoided damages in $ billions. The buyout costs $4 billion today.

Year (Y)	Avoided damages	PV at 7%	PV at 3%
40	$15		
80	$40		
Total	$55		

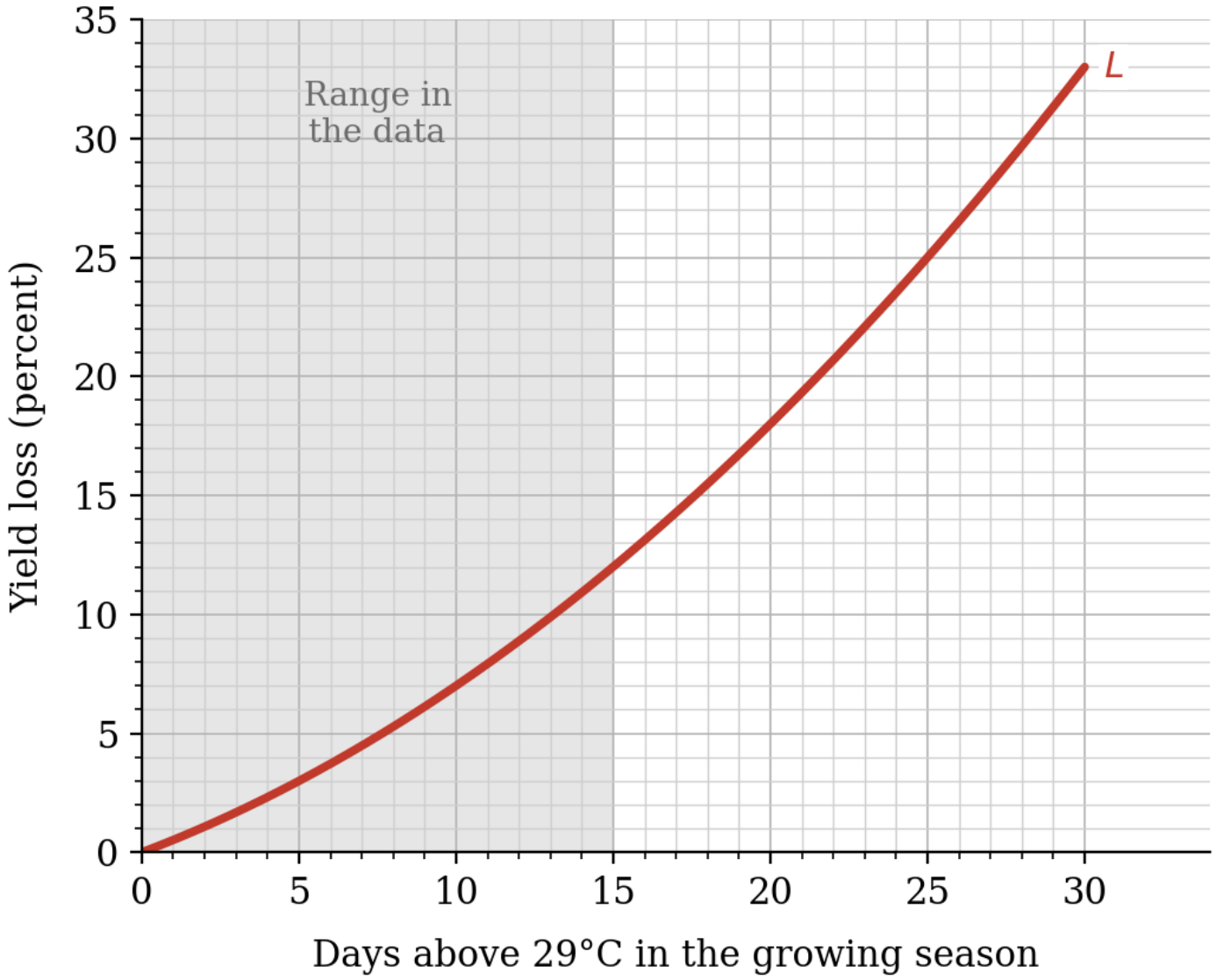
Practice problem 2

Damage functions and adaptation

- Researchers estimate how extreme heat reduces a county's corn yield:
- **$L = 0.5D + 0.02D^2$**
- D = growing-season days above 29°C. L = % yield loss. Their data cover seasons with 0 to 15 hot days.
- Climate change raises hot days from 10 today to 25 by 2075. A season with no hot days has corn yields worth $90M.

  - (a) What is yield loss today and in 2075?
  - (b) What is dollar value of the added loss each year?
  - (c) A linear fit to the same data gives L = 0.8D. Repeat (a) and (b). What explains the gap?
  - (d) Farmers adapt by irrigating: $L_I = 0.4D + 0.01D^2$, at $3M per year. Total climate damages per year?

Practice problem 2



$L = 0.5D + 0.02D^2$

Data: 0 to 15 hot days

Today 10 days. 2075: 25 days.

Full yield: 20m bushels × \$4.50

(a) Loss today and in 2075.

(b) Dollar value of the added loss.

(c) Repeat with L = 0.8D. Why the gap?

(d) $L_I = 0.4D + 0.01D^2$, \$3m per year. Total damages?

Practice problem 3

The VSL and the SCC

- A city offers cooling centers during heat waves. Each resident's risk of dying from heat falls by 1 in 10,000 per year. The average resident would pay $1,100 per year.
- A climate model finds that 1M more tons of CO_2 emitted today cause 4 extra heat deaths worldwide, all 50 years from now. Use r = 2%.

  - (a) What VSL does the survey imply?
  - (b) What is the nominal (i.e., not discount) value of the deaths per ton of CO_2?
  - (c) What is the present value per ton (i.e., the mortality part of the SCC)?
  - (d) 1 of the 4 deaths is in the US. What is the US share of the mortality-only SCC? Why not set a US carbon price at that number?

Practice problem 3

Pay $1,100 for a 1 in 10,000 lower heat risk. 4 deaths per million tons in year 50. Rate 2%.

Step	Value
(a) Value of a statistical life	
(b) Value of deaths per ton, undiscounted	
(c) Discount factor, 1.02^{50}	
(c) Present value per ton	
(d) US share per ton	